PUNITH KM

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| B.Tech CSE — Artificial Intelligence | 2027 Graduate

PROFESSIONAL SUMMARY

Motivated Computer Science fresher (2027 graduate) specializing in Artificial Intelligence with strong programming fundamentals in Python, Java, and C. Built real-time distributed systems, full-stack applications, and computer vision pipelines independently. Experienced in system design, fault-tolerant architecture, and low-latency communication protocols. Seeking a Software Engineer role to contribute to scalable, production-grade software development.

EDUCATION

BTech — Computer Science Engineering (Artificial Intelligence)
Parul University | 2023 – 2027 | CGPA: 8.04

EXPERIENCE

Self-Employed | Software Developer *2024 – Present*

- Developed real-time system pipelines handling concurrent task execution and fault recovery across networked nodes
- Implemented low-latency processing pipelines for autonomous command execution in peer-to-peer networked systems
- Constructed a gesture-recognition interface using OpenCV and MediaPipe achieving sub-100ms command response
- Delivered full-stack applications using Python and JavaScript with REST-based backend architecture
- Maintained version-controlled codebases on GitHub with modular, well-documented software design

TECHNICAL SKILLS

Programming Languages: Python, Java, C, JavaScript
Systems & Networking: UDP protocol, peer-to-peer architecture, heartbeat monitoring, leader election, fault-tolerant systems, REST APIs
Core Concepts: Data Structures, Algorithms, Object-Oriented Programming, Operating Systems
Web Technologies: HTML, CSS, JavaScript, Streamlit
Computer Vision: OpenCV, MediaPipe, real-time image processing
Databases: MySQL, MongoDB
Tools: Git, GitHub, VS Code, Postman, Linux

PROJECTS

Fault-Tolerant Distributed System — Drone Coordination Network | *Python · UDP · OpenCV · Streamlit* | *Completed*

- Architected a fault-tolerant distributed system managing 5+ concurrent nodes with dynamic leader election and automatic failover
- Implemented UDP-based peer-to-peer networking with a heartbeat protocol for real-time node health monitoring
- Engineered a sub-100ms command processing pipeline over UDP for low-latency autonomous node coordination
- Launched a live monitoring dashboard providing real-time visibility into node health, commands, and system state

Tourist Guide Recommendation System | *HTML · CSS · JavaScript · Python* | *Completed*

- Built a full-stack web application covering 50+ tourist locations with dynamic, real-time filtering logic
- Connected a Python backend with a JavaScript frontend, reducing average page load time by 30%
- Implemented location-aware recommendation logic with structured data handling and responsive UI design

Credit Card Fraud Detection Pipeline | *Python · Scikit-learn · XGBoost · Tableau* | *Completed*

- Developed an end-to-end data processing and ML pipeline on 284,000+ transactions achieving 97.8% accuracy and 91% recall
- Engineered 10+ features and benchmarked 3 models using ROC-AUC and F1-score, reducing false positive rate by 23%
- Produced an interactive Tableau dashboard with anomaly trend visualization for stakeholder reporting

CERTIFICATIONS & ACHIEVEMENTS

- Machine Learning Workshop — IIT Bombay Techfest (national-level technical festival)
- Agentic AI Workshop — Advanced AI systems & multi-agent orchestration
- Data Science Workshop — Data Cleaning, Preprocessing & Feature Engineering
- Prompt Engineering — Advanced techniques for large language models
- National-level SIH participant | Greenbottle sustainable product concept presented to industry judges

KEY STRENGTHS

Systems Thinking | Problem Solving | Analytical Thinking | Qick Thought